

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: Cloud Computing
and Big Data
Analytics

COURSE CODE: CSA5112

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)*

Dave's Taxonomy: Manipulation
(Level 2) Question Count: 5

**Total
Marks: 50**

Code of Conduct

I T.GOKULNATH /192521139 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the student: T.GOKULNATH

Assessment Tasks

Analytical Problem

Solving

1. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.
2. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.

Analytical Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 – Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

ANSWERS

1. **A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.**

Introduction

Cloud computing has revolutionized the way organizations deploy and manage applications by enabling multiple workloads to share the same physical infrastructure. Organizations such as banks, hospitals, educational institutions, government agencies, and multinational corporations increasingly rely on cloud platforms because they reduce hardware costs, simplify management, and improve scalability. However, one of the greatest challenges in cloud computing is ensuring that different organizations or departments sharing the same physical hardware remain completely isolated from one another.

Consider a cloud provider hosting three highly sensitive workloads:

- **Finance applications**, containing confidential banking transactions and payment records.
- **Healthcare applications**, storing electronic medical records (EMRs), patient histories, prescriptions, and diagnostic reports.
- **Student portals**, maintaining academic records, examination results, attendance, and personal information.

Although all three workloads may run on the same physical server, they must operate as though each has its own dedicated hardware. A security breach affecting one workload must never compromise the others. This level of separation is made possible through **virtualization architecture**, which creates isolated Virtual Machines (VMs) on shared hardware.

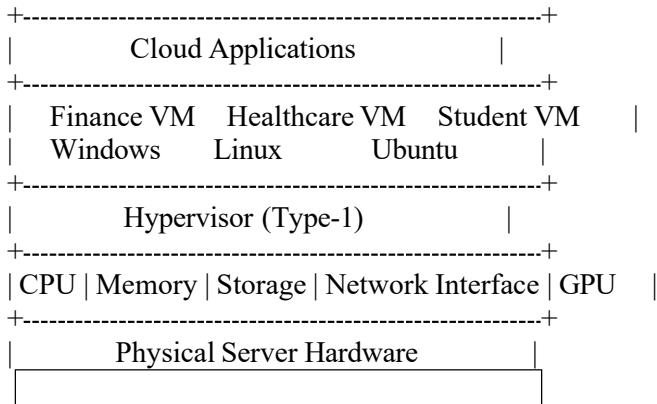
Virtualization architecture is the foundation of Infrastructure as a Service (IaaS) platforms such as VMware vSphere, Microsoft Hyper-V, KVM, Xen, and cloud providers including AWS EC2, Microsoft Azure Virtual Machines, and Google Compute Engine. It allows efficient resource utilization while maintaining security, scalability, flexibility, and fault isolation.

Virtualization Architecture

Virtualization architecture refers to the technology that enables multiple virtual computers to run simultaneously on a single physical machine by abstracting hardware resources.

Instead of directly running one operating system on hardware, a software layer known as the **hypervisor** manages the physical resources and allocates them to multiple virtual machines.

Basic Architecture



Each virtual machine behaves like an independent computer with its own:

- Operating system
- CPU allocation
- RAM allocation
- Virtual disk
- Network adapter
- Security policies
- File system

Even though all VMs share the same hardware, they remain logically separated.

Components of Virtualization Architecture

1. Physical Hardware

The physical server consists of:

- Multi-core processors
- RAM
- SSD/HDD storage
- Network Interface Cards (NICs)
- Power supply
- GPU (optional)

The hardware provides computing resources that are shared among virtual machines.

2. Hypervisor

The hypervisor is the most important component.

It creates and manages virtual machines while ensuring isolation.

Responsibilities

- Resource allocation
- CPU scheduling
- Memory management
- Virtual networking
- Storage virtualization
- Security enforcement
- VM monitoring

Examples include:

- VMware ESXi
- Microsoft Hyper-V
- KVM
- Xen

Types of Hypervisors

Type-1 (Bare Metal)

Installed directly on hardware.

Applications



Guest OS



Hypervisor



Hardware

Advantages

- High performance
- Better security
- Enterprise cloud support
- Lower overhead

Examples

- VMware ESXi
- Hyper-V Server
- Xen

Type-2 Hypervisor

Runs on top of a host operating system.

Applications



Guest OS



Hypervisor



Host Operating System



Hardware

Examples

- Oracle VirtualBox
- VMware Workstation

Used mainly for development and testing.

How Virtualization Supports Isolation

Isolation means one virtual machine cannot interfere with another VM.

This isolation is achieved using several mechanisms.

1. CPU Isolation

Each VM receives dedicated virtual CPUs (vCPUs).

The hypervisor schedules CPU time fairly.

Example
16-Core Processor

↓

Finance VM → 4 cores

Healthcare VM → 6 cores

Student Portal → 6 cores

Even if the healthcare VM consumes heavy CPU resources, it cannot directly access the finance VM's CPU registers or execution state.

Benefits:

- Fair scheduling
- No CPU interference
- Predictable performance

2. Memory Isolation

Memory is isolated using virtual memory management.

Each VM believes it owns its allocated RAM.

Example
64 GB RAM

↓

Finance VM → 20 GB

Healthcare VM → 24 GB

Student Portal → 20 GB

The hypervisor uses:

- Shadow page tables
- Extended Page Tables (EPT)
- Nested Page Tables (NPT)

These ensure that:

- Finance VM cannot read Healthcare VM memory.
- Healthcare VM cannot modify Student VM memory.
- Student Portal cannot access Finance VM RAM.

Without virtualization, confidential information such as passwords or encryption keys could leak across applications.

3. Storage Isolation

Every VM has its own virtual hard disk.

Example
Physical SSD

↓

Finance.vmdk

Healthcare.vmdk

Student.vmdk

Each virtual disk:

- Has separate permissions
- Uses independent file systems
- Can be encrypted individually
- Can be backed up separately

Benefits:

- Data confidentiality
- Easy disaster recovery
- Independent snapshots

4. Network Isolation

Networking is virtualized using:

- Virtual switches
- VLANs
- VXLAN
- Software Defined Networking (SDN)

Example

Virtual Switch

|

+-----+

Finance VLAN 10

Healthcare VLAN 20

Student VLAN 30

Even though packets travel through the same physical NIC, VLAN tagging prevents unauthorized communication. Firewall rules further restrict traffic.

Example

Finance VM

↓

Only database server allowed

↓

Healthcare VM blocked

↓

Student Portal blocked

5. Device Isolation

Virtual machines receive virtual hardware.

Examples

- Virtual NIC
- Virtual Disk Controller
- Virtual USB
- Virtual GPU

Direct hardware access is prevented unless specifically configured using PCI passthrough.

6. Process Isolation

Every operating system runs independently.

Finance Windows Server

↓

Own processes

↓

Own services

↓

Own kernel

Healthcare Linux

↓

Own kernel

↓

Own process table

7. Fault Isolation

Suppose:

Healthcare application crashes.

The result:

✓ Finance VM continues working.

✓ Student Portal remains online.

Only Healthcare VM restarts.

This greatly improves availability.

Security Features Used for Isolation

Modern hypervisors implement numerous security mechanisms.

Secure Boot

Ensures only trusted operating systems are loaded.

Prevents rootkits.

Trusted Platform Module (TPM)

Provides hardware-based encryption.

Stores cryptographic keys securely.

VM Encryption

Each VM disk is encrypted.

Even if storage is stolen,

Data remains unreadable.

Snapshots

Hypervisors create VM snapshots.

Benefits

- Backup
 - Rollback
 - Malware recovery
 - Testing
-

Role-Based Access Control (RBAC)

Administrators receive different privileges.

Example

Role	Permission
Cloud Admin	Full Access
Finance Admin	Finance VM Only
Hospital Admin	Healthcare VM Only
Student Admin	Student VM Only

This prevents accidental modifications.

Isolation Example

Suppose the finance application processes online banking.

Healthcare stores MRI reports.

Student Portal hosts examination results.

A ransomware attack infects Student VM.

Without virtualization:

Entire server compromised.

With virtualization:

Student VM isolated.

Finance unaffected.

Healthcare unaffected.

Administrator restores Student VM snapshot.

Major Attack Surface 1 – Hypervisor Attack (VM Escape)

One of the most dangerous threats in virtualization is **VM Escape**.

Definition

VM Escape occurs when malicious code running inside a virtual machine exploits a vulnerability in the hypervisor and escapes the VM boundary to execute code on the host system.

Once successful, the attacker may gain control over:

- Host operating system (if applicable)
- Hypervisor
- Other virtual machines
- Storage resources
- Network configuration

Attack Diagram

Attacker



Compromised Student VM



Hypervisor Vulnerability



Hypervisor Control



Finance VM



Healthcare VM

Consequences

- Theft of banking information
- Exposure of medical records
- Leakage of student databases
- Service disruption
- Complete cloud compromise

Mitigation

- Keep hypervisor patched.
- Use Type-1 hypervisors.
- Disable unnecessary services.
- Enable Secure Boot.
- Apply least-privilege access.
- Perform regular vulnerability scanning.
- Use hardware-assisted virtualization (Intel VT-x, AMD-V).

Major Attack Surface 2 – Shared Resource Side-Channel Attacks

Although VMs are isolated logically, they still share some hardware components such as:

- CPU cache
- Memory bus
- Branch predictor
- Shared processors

Attackers can exploit timing differences in shared hardware to infer sensitive information. These are known as **side-**

channel attacks.

Examples

- Cache timing attacks
- Spectre
- Meltdown
- Flush+Reload
- Prime+Probe

Attack Flow

Finance VM

↓

Uses CPU Cache

↓

Attacker VM Measures Cache Timing

↓

Infers Secret Keys

↓

Data Leakage

Consequences

- Exposure of encryption keys
- Password leakage
- Cryptographic attacks
- Information disclosure

Mitigation

- Apply CPU microcode updates.
- Enable hardware security features.
- Separate sensitive workloads onto dedicated hosts.
- Disable simultaneous multithreading (SMT) where appropriate.
- Use cache partitioning techniques.
- Continuously monitor for abnormal behavior.

Additional Security Best Practices

To strengthen workload isolation further, cloud providers should:

- Encrypt data at rest and in transit.
- Use multi-factor authentication (MFA) for administrators.
- Implement network micro-segmentation.
- Conduct regular penetration testing.
- Monitor logs using SIEM tools.
- Apply least-privilege principles.
- Use intrusion detection and prevention systems (IDS/IPS).
- Schedule periodic security audits and compliance checks (e.g., PCI DSS for finance, HIPAA-like controls for healthcare where applicable).

Advantages of Virtualization-Based Isolation

Feature	Benefit
Resource Sharing	Better hardware utilization
VM Isolation	Prevents cross-VM interference
Scalability	Easy to create new VMs
Snapshots	Fast recovery after failures
Security	Strong workload separation

Feature	Benefit
Live Migration	Maintenance without downtime
Disaster Recovery	Simplified backup and restoration
Flexibility	Supports multiple operating systems on one host

Conclusion

Virtualization architecture enables finance, healthcare, and student portal applications to safely coexist on the same physical hardware while maintaining strong logical separation. The hypervisor isolates CPU, memory, storage, networking, devices, and operating systems, ensuring that each workload functions independently. This approach improves hardware utilization, scalability, and operational efficiency while reducing infrastructure costs. However, virtualization is not immune to security risks. Two of the most critical attack surfaces are **hypervisor attacks (VM Escape)** and **shared-resource side-channel attacks**, both of which can undermine isolation if left unaddressed. By combining secure hypervisors, regular patching, encryption, network segmentation, hardware-assisted virtualization, and continuous monitoring.

2. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.

Performance Analysis of VM Host Utilization

Introduction

Virtualization technology enables multiple virtual machines (VMs) to share the same physical hardware while maintaining isolation and efficient resource utilization. However, as more VMs are deployed on a host, the underlying hardware resources such as CPU, memory, storage, and network may become heavily utilized. Excessive resource usage results in performance degradation, longer response times, reduced throughput, application failures, and poor user experience.

The given VM host utilization snapshot indicates that the host server is experiencing high resource utilization:

- CPU Utilization = 92%
- Memory Utilization = 88%
- Storage I/O Latency = 35 ms
- Average VM Boot Time = 170 seconds

These values suggest that the virtualization host is under heavy load and may not have enough available resources to efficiently support all running virtual machines. A systematic performance analysis is necessary to identify the root causes of these bottlenecks and recommend corrective actions to restore optimal performance.

Snapshot Summary

Resource	Observed Value	Status
CPU	92%	Critical
Memory	88%	Very High
Storage Latency	35 ms	Poor
VM Boot Time	170 s	Very Slow

Normally,

Resource	Recommended
CPU	Below 70%
Memory	Below 75%
Storage Latency	Below 10 ms
VM Boot Time	30–60 seconds

The observed values clearly indicate that the host is overloaded.

Understanding Resource Bottlenecks

A bottleneck is any resource whose limited capacity restricts the overall performance of the system.

Example:
Application

↓

Virtual Machine

↓

Hypervisor

↓

CPU
RAM
Storage
Network

If any one component becomes overloaded, every VM running on the host is affected.

CPU Bottleneck Analysis

Given Value

CPU Utilization = 92%

This is extremely high.

Normally CPU utilization above **85%** indicates that processors are saturated.

At 92%, the hypervisor has very little free CPU capacity remaining.

Causes of High CPU Utilization

Several reasons may contribute to this condition.

1. Too Many Running Virtual Machines

The host may be running more VMs than its processors can efficiently support.

Example

16-core server

↓

30 virtual machines

↓

Each VM requesting multiple vCPUs

↓

CPU contention

2. CPU Overcommitment

Hypervisors often assign more virtual CPUs than physical CPUs.

Example

Physical CPU

16 cores

Allocated

40 virtual CPUs

The hypervisor continuously switches between VMs.

This causes

- Context switching
- Scheduling delays
- Increased CPU wait time

3. High Workload Applications

Some applications naturally consume high CPU.

Examples

- Database servers
- AI workloads
- Video rendering

- Data analytics
- Scientific simulations

If several CPU-intensive applications run simultaneously, CPU usage increases rapidly.

4. Background Hypervisor Processes

The hypervisor itself requires CPU resources for

- Memory management
- VM scheduling
- Live migration
- Snapshots
- Backup operations

Heavy management tasks further increase CPU utilization.

Effects of High CPU Usage

High CPU utilization affects the entire virtualized environment.

Effects include

- Slow applications
- Increased response time
- VM freezing
- Longer login times
- High CPU Ready Time
- Delayed scheduling
- Reduced throughput
- Poor user experience

Example

CPU Busy

↓

VM waits

↓

Applications slow

↓

Users experience delay

Corrective Actions for CPU Bottleneck

1. Live Migration

Move heavily loaded VMs to another host.

Benefits

- Reduces CPU load
 - No downtime
 - Better load balancing
-

2. Load Balancing

Use

- VMware DRS
- Hyper-V Dynamic Optimization
- Kubernetes scheduling (for containers)

These automatically distribute workloads.

3. Upgrade CPU

Install

- More processor cores
- Faster processors

- Additional sockets

4. Reduce CPU Overcommitment

Avoid assigning excessive virtual CPUs.

Example

Instead of

VM = 8 vCPUs

Assign

VM = 2–4 vCPUs

unless required.

5. Shut Down Idle VMs

Unused virtual machines still consume resources.

Powering them off immediately reduces CPU load.

Memory Bottleneck Analysis

Given Value

Memory Utilization

88%

This indicates that RAM is almost exhausted.

The hypervisor has very little free memory remaining.

Causes of High Memory Usage

Too Many VMs

Every virtual machine reserves RAM.

Example

Finance VM

16 GB

Healthcare VM

16 GB

Student VM

16 GB

Database VM

32 GB

Soon the physical memory becomes fully utilized.

Memory Leaks

Applications sometimes fail to release unused memory.

This gradually consumes available RAM.

Large Databases

Database caching requires significant memory.

Examples

Oracle

SQL Server

MongoDB

PostgreSQL

Poor Memory Allocation

Some VMs may have excessive RAM assigned but use only a small fraction.

Example

Allocated

16 GB

Actually Used

3 GB

Remaining memory remains unavailable to other VMs.

Effects of Memory Bottleneck

High memory utilization results in

- Memory swapping
 - Ballooning
 - Slow applications
 - Disk paging
 - VM crashes
 - Host instability
-

Memory Swapping

When RAM is exhausted,
the hypervisor stores memory pages on disk.

RAM

↓

Full

↓

Disk Swap

↓

Much slower performance

Since disks are much slower than RAM, application performance decreases significantly.

Corrective Actions for Memory Bottleneck

Increase RAM

Upgrade physical memory.

Example

64 GB

↓

128 GB

Dynamic Memory Allocation

Use

- VMware Ballooning
- Hyper-V Dynamic Memory

Unused memory can be reassigned automatically.

Remove Idle VMs

Unused machines consume valuable RAM.

Delete or power off inactive VMs.

Optimize Applications

Reduce unnecessary services

Close unused processes

Tune database cache size

Monitor Memory Usage

Use monitoring software

Examples

- VMware vCenter
 - Prometheus
 - Grafana
 - Nagios
 - Zabbix
-

Storage Bottleneck Analysis

Given Value

Storage Latency

35 milliseconds

This is significantly higher than recommended values.

Typical storage latency

SSD

1–5 ms

Enterprise SAN

5–10 ms

35 ms indicates severe storage congestion.

Causes

Too Many Simultaneous Disk Requests

Multiple VMs may access storage simultaneously.

Example

Finance Database

↓

Healthcare Database

↓

Student Portal

↓

Backup Software

↓

Same SSD Array

↓

Storage Queue Builds

Slow HDD Storage

Traditional hard disks cannot process requests as quickly as SSDs.

Backup Operations

Nightly backups create heavy read/write traffic.

Snapshot Chains

Numerous VM snapshots increase storage overhead and latency.

Insufficient RAID Configuration

Poor RAID selection reduces storage performance.

Example

RAID 5

↓

Higher write latency

RAID 10

↓

Better virtualization performance

Effects of High Storage Latency

- Slow application loading
 - Long boot times
 - Slow database queries
 - Delayed file access
 - Increased VM response time
-

Corrective Actions

Upgrade Storage

Replace HDD

↓

Enterprise SSD

or

NVMe Storage

Separate Workloads

Store databases on dedicated storage.

Keep backups on different disks.

Remove Old Snapshots

Long snapshot chains increase disk overhead.

Delete unnecessary snapshots regularly.

Optimize RAID

Use RAID 10 for virtualization environments requiring high I/O performance.

Enable Storage Tiering

Frequently accessed ("hot") data stays on SSDs.

Less frequently accessed ("cold") data moves to HDDs.

This improves overall storage efficiency.

VM Boot Time Analysis

Given Value

Average Boot Time

170 seconds

Normal boot time

30–60 seconds

170 seconds is nearly **3 minutes**, indicating significant performance degradation.

Causes

- CPU contention
 - Memory pressure
 - High storage latency
 - Large startup applications
 - Antivirus scanning during boot
 - Too many startup services
-

Effects

- Delayed recovery after failures
 - Longer maintenance windows
 - Reduced productivity
 - Poor availability of services
-

Corrective Actions

- Store VM images on SSD/NVMe.
 - Reduce unnecessary startup services.
 - Increase CPU and RAM availability.
 - Optimize guest operating systems.
 - Use VM templates and linked clones for faster deployment.
 - Schedule antivirus scans outside boot periods.
-

Relationship Between the Bottlenecks

These issues are interconnected rather than isolated.

CPU 92%



VMs wait for processing



Applications remain active longer



Memory stays occupied



Memory reaches 88%



Swapping to disk begins
↓
Storage latency increases to 35 ms
↓
VM boot time rises to 170 seconds
A bottleneck in one resource often cascades into the others, amplifying overall performance degradation.

Overall Corrective Strategy

Issue	Recommended Action
CPU Saturation	Live migration, reduce CPU overcommitment, add processors
High Memory Usage	Upgrade RAM, dynamic memory allocation, remove idle VMs
Storage Latency	Upgrade to SSD/NVMe, optimize RAID, remove old snapshots
Slow Boot Time	Optimize startup, improve storage performance, reduce contention

Best Practices for Maintaining VM Host Performance

- Continuously monitor CPU, memory, storage, and network utilization.
- Keep average CPU utilization below **70%** to allow headroom for spikes.
- Maintain memory utilization below **75%** to avoid swapping.
- Use SSD or NVMe storage for production virtual machines.
- Balance workloads across multiple hosts using clustering and live migration.
- Avoid excessive CPU and memory overcommitment.
- Remove inactive VMs, unused snapshots, and orphaned virtual disks.
- Schedule backups and antivirus scans during off-peak hours.
- Apply regular hypervisor updates and firmware patches.
- Implement automated alerts for resource thresholds.

Conclusion

The given utilization snapshot clearly indicates that the virtualization host is under heavy stress. A **CPU utilization of 92%** signifies processor saturation, while **memory utilization of 88%** leaves little available RAM and increases the likelihood of memory swapping. A **storage I/O latency of 35 ms** points to severe storage contention, likely caused by excessive disk requests, inefficient storage architecture, or snapshot accumulation. Consequently, the **average VM boot time of 170 seconds** is far above acceptable levels, reflecting the combined impact of CPU, memory, and storage bottlenecks.

To restore optimal performance, administrators should adopt a comprehensive optimization strategy that includes upgrading hardware resources, balancing workloads across hosts, reducing resource overcommitment, optimizing storage with SSD/NVMe devices, implementing dynamic memory management, and continuously monitoring host performance. By addressing these bottlenecks proactively, cloud providers can ensure improved VM responsiveness, higher availability, enhanced scalability, and a better overall user experience.